

Arm® Cortex®-M55
32-bit Microcontroller

NuMicro® Family
M5531 Series BSP
Revision History

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Revision 3.01.005 (Released 2026-07-24)

- Add SBOM.
- CMSIS upgrades to 6.3.0 and adds CMSIS-Compiler 2.2.0.
- Startup
 - Reset_Handler_Prelinit fixes incorrect timeout check.
- Standard driver
 - FMC: FMC_WriteMultiple supports resume for timeout recovery.
 - I2S: I2S_ORDER_AT_LSB and I2S_ORDER_AT_MSB fix wrong definition.
 - LTPWM: LTPWM_ENABLE_OUTPUT fixes disable functionality.
- CMSIS driver
 - USB: Support HSUSB DMA transfer.
- Library
 - UsbHostLib supports UAC 2.0.
- ThirdParty
 - FatFs: Upgrade to R0.16.
- Sample code
 - All sample code projects add no-unaligned-access compiler flag.
 - DMIC_I2S_Play and DMIC_WAVRecorder: Add 10 ms DMIC turn-on time.
 - EPWM: EPWM_EnableFaultBrake removes support of multiple brake sources.
 - FMC_XOM, XOMLib, and XOMLibDemo: Refactor for dual bank.
 - HSUSB mass-storage sample codes: Use memory buffer to replace fixed address.
 - HSUSB_MassStorage_ScatterGather: Align buffer for Cache/DMA coherence.
 - HSUSB_RNDIS: Improve performance of bulk transfer with DMA.
 - HSUSBH_USBH_AudioClass, HSUSBH_USBH_UAC_HID, and HSUSBH_USBH_UAC_Loopback: Support UAC 2.0.
 - I2S_Codec_PDMA: Support 24-bit data width.
 - ISP_DFU: Remove dfu-util.
 - LwIP_MQTT: Fix semaphore double-take issue.
 - SDH_FATFS: Improve SD card hot-plug response time.
- Tool
 - HIDTransferTest
 - Move from USB sample codes.
 - Add Control transfer.
 - mcuboot: Upgrade to v2.4.0 with imgtool.

Revision 3.01.004 (Released 2026-03-26)

- Peripheral register
 - fmc_reg.h: Define FMC_ISPSTS_INTFLAG_Pos based on secure state.
- Standard driver
 - EADC: EADC_Open removes EADC_Calibration call.
 - EPWM: EPWM_EnableFaultBrake supports multiple brake sources.
 - I3C: Update for new I3C sample codes.
 - PMC: PMC_SetPowerDownMode adds NPD3 power-down mode limitation.
 - RNG: Remove SysTick dependency.
 - SPIM: Improve stability of quad I/O transfer mode and 4-byte address mode.
- CMSIS driver
 - Flash: Support PDMA and non-blocking mode operations.
- Library

- UsbHostLib
 - Support still image capture.
 - Reduce memory usage and fix device disconnect flow.
- xmodem: Fix 128-byte unalignment send issue.
- Sample code
 - CANFD_TxRxINT: Fix wrong bitmask in CANFD00_IRQHandler.
 - EADC samples: Fit max 90 MHz EADC clock frequency.
 - HSUSBH_USBH_UVC: Support video still image capture.
 - I3C_Controller: Replace samples of I3C_Master_IBI, I3C_MasterRW, I3C_MasterRW_PDMA and I3C_SecondaryMaster.
 - I3C_Target: Replace samples of I3C_Slave_HotJoin, I3C_Slave_IBI, I3C_Slave_Wakeup, I3C_SlaveRW, I3C_SlaveRW_PDMA and I3C_SecondaryMaster.
 - ISP samples
 - Fix flash select setting in flash download of Keil project.
 - Fix the issue of “Reset and Run” command not booted from APROM.
 - QSPI_DualMode_Flash, QSPI_QualMode_Flash, SPI_Flash: Improve performance.
 - RNG_EntropyPoll, RNG_Random: Remove SysTick dependency.
 - SPIM samples: Update trim procedure to reduce times to erase or program Flash.
 - Remove sample codes not applied to NuMaker-M5531 board.

Revision 3.01.003 (Released 2025-10-29)

- Peripheral register
 - eadc_reg.h: Remove some redundant bit-field macros.
- Standard driver
 - retarget.c: Improve compatibility with CMSIS RTE.
 - CLK: Remove CLK_APLLCTL_APLLSRC_HXT_DIV2.
 - LPPDMA: Redefine LPPDMA_INT_TRANS_DONE and LPPDMA_INT_EMPTY to support multi-bit mask.
 - PDMA
 - PDMA_INT_TRANS_DONE, PDMA_INT_EMPTY, PDMA_INT_TIMEOUT, PDMA_DisableInt, PDMA_EnableInt: Support multi-bit mask.
 - PDMA_INT_ALIGN: Add.
 - SPIM: Remove voltage raise for HyperBus device and octal SPI flash.
- CMSIS driver
 - RTE_Device.h: Improve compatibility with CMSIS packs.
 - I2C: Master adds I3C IP support.
 - USB: Add HSUSB IP support.
 - VIO: Add NuMaker board support.
- Library
 - UsbHostLib: Improve robustness of memory management and EHCI interrupt handling.
- Sample code
 - DMIC_I2S_Play, I2S_Codec_PDMA, I2S_MP3Player, I2S_WAVEPlaye: Fix cache line alignment issues of I2S PDMA buffers.
 - EADC_AverageCMP: Add.
 - MP3_Recorder: Add USB storage device support.

- VS Code projects upgrade to use GCC 14.3.1.

Revision 3.01.002 (Released 2025-8-8)

- Peripheral register
 - canfd_reg.h
 - CANFD_RXF0A_F0AI_Msk: Replace CANFD_RXF0A_F0AI_Msk.
 - CANFD_RXF0A_F0AI_Pos: Replace CANFD_RXF0A_F0AI_Pos.
 - hsusbd_reg.h: Fix definitions of HSUSBD_EPBUFSTART_SADDR_Msk and HSUSBD_EPBUFSTART_SADDR_Pos.
- Startup
 - SysTick vector on flash uses Default_Handler.
- Standard driver
 - CANFD: Remove unsupported UTSU function.
 - HSUSBD: Add HSUSBD_DISABLE_HS_HANDSHAKE and HSUSBD_ENABLE_HS_HANDSHAKE.
 - SPI: Add SPI_DISABLE_3WIRE_MODE and SPI_ENABLE_3WIRE_MODE.
 - SPIM
 - Raise voltage to 1.2V for HyperBus device and octal SPI flash.
 - Support Infineon octal SPI flash.
 - SYS
 - SYS_GPC_MFP0_PC2MFP_UTCPD0_CCDB2: Replace SYS_GPC_MFP0_PC2MFP_UTCPD0_CCDB1.
 - SYS_GPC_MFP0_PC3MFP_UTCPD0_CCDB1: Replace SYS_GPC_MFP0_PC3MFP_UTCPD0_CCDB2.
 - SYS_ResetModule: Add SPIM clock setting handling.
- CMSIS driver
 - CAN, ETH_MAC, Flash, GPIO, I2C, MCI, SAI, SPI, USART, USBD, USBH: Add.
- Library
 - CryptoAccelerator: Improve cache coherence.
 - JpegAcceleratorLib: Improve performance.
 - UsbHostLib: Add UVC support.
- Sample code
 - AWF_GSensor_Wakeup: Use I/O buffer to replace fixed address.
 - DMIC_VAD_Wakeup: Increase VAD power threshold.
 - HSUSBD_Audio20_Codec and HSUSBD_Audio20_Headset: Fix no sound issue.
 - HSUSBD_HID_Mouse: Improve LPM stability.
 - HSUSBD_RNDIS: Add.
 - HSUSBD_Video_CAM: Fix UVC image noise issue.
 - HSUSBH_USBH_UVC: Add.
 - I3C samples: I3C pins enable Schmitt trigger.
 - ISP_DFU_20 and ISP_HID_20: Fix USB high-speed issue.
 - ISP_I2C and ISP_SPI: Fix potential command failure.
 - KS_AESKey: Improve cache coherence.
 - RTC_TimeAndTick: Remove because almost the same as RTC_Time_Display.
 - SecureApplication samples: VS Code projects require Python 3.12 at least for post-build.

- SPIM_HYPER_ExeInHRAM: Fix I/O buffer alignment and GCC project execution failure.
- SPIM_HYPER_RW_MemMap: Fix I/O buffer alignment.
- USB_D samples: Improve stability and performance.
- VS Code projects update for Nuvoton NuMicro Cortex-M Pack.

Revision 3.01.001 (Released 2025-3-26)

- Initial release.

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